INCH-POUND
MIL-DTL-3650B
w/AMENDMENT 1
14 November 2022
SUPERSEDING
MIL-DTL-3650B
25 November 2003

DETAIL SPECIFICATION

CONNECTORS, COAXIAL, RADIO FREQUENCY, SERIES LC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

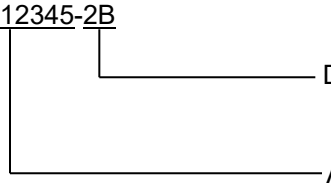
1.1 Scope. This specification covers weatherproof, series LC, radio frequency, coaxial connectors having a nominal impedance of 50 ohms (see 6.1 and 6.2).

1.2 Classification.

1.2.1 Type designation. The type designation of connectors is derived from the AN nomenclature system specified in MIL-STD-196 (see 3.1 and 6.2.)

1.2.2 Part or Identifying Number (PIN). The PIN consists of the applicable "MS" PIN.

Example: MS12345-2B



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, Attn: VAI, P.O. Box 3990 East Broad Street, Columbus, Ohio 43216-5000 or emailed to RFConnector@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil/.

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FEDERAL SPECIFICATIONS

L-P-516	-	Plastic Sheet and Plastic Rod, Thermosetting, Cast.
O-F-499	-	Flux, Brazing, (Silver Brazing Filler Metal, Low Melting Point).
QQ-B-654	-	Brazing Alloys, Silver.

FEDERAL STANDARDS

FED-STD-H28	-	Screw-Thread Standards for Federal Services.

COMMERICAL ITEM DESCRIPTIONS

A-A-59588	-	Rubber, Silicone.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-17/65	- Cables, Radio Frequency, Flexible Coaxial, 50 OHMS, Unarmored M17/65-RG165, and Armored, M17/65-RG166
MIL-DTL-28875	- Amplifiers, Radio-Frequency and Microwave, Solid-State, General Specification for
MS91604	- Connector, Plug, Electrical, Series LC, UG-154A/U
MS91610	- Connector, Receptacle, Electrical, Series LC, UG-352B/U
MS91618	- Adapter, Connector, Series LC, UG-287B/U

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	- Electronic and Electrical Component Parts.
MIL-STD-202-101	- Method 101, Salt Atmosphere (Corrosion)
MIL-STD-202-301	- Method 301, Dielectric Withstanding Voltage
MIL-STD-348	- Radio Frequency Connector Interfaces for MIL-DTL-3643, MIL-DTL-3650, MIL-DTL-3655, MIL-DTL-25516, MIL-PRF-31031, MIL-PRF-39012, MIL-PRF-49142, MIL-PRF-55339, MIL-DTL-83517
MIL-STD-1285	- Marking of Electrical and Electronic Parts
MIL-STD-1353	- Electrical Connectors, Plug-in Sockets and Associated Hardware, Selection and Use of

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B16/B16M	- Rod, Brass, Free-Cutting, Bar and Shapes For Use in Screw Machines.
ASTM B36/B36M	- Plate, Brass Sheet, Strip and Rolled Bar.
ASTM B121/B121M	- Plate, Leaded Brass Sheet, Strip and Rolled Bar.
ASTM B124/B124M	- Copper and Copper Alloy Forging Rod, Bar and Shapes.
ASTM B139/B139M	- Rod, Phosphor Bronze Bar and Shapes.
ASTM B194	- Copper Beryllium Alloy Plate, Sheet, Strip and Rolled Bar.
ASTM B196/B196M	- Rod and Bar, Copper Beryllium Alloy.
ASTM B197/B197M	- Wire, Alloy Copper-Beryllium.
ASTM B700	- Electrodeposited Coatings of Silver for Engineering Uses.
ASTM D2000	- Rubber Products in Automotive Applications.
ASTM D4976	- Polyethylene Plastics Molding and Extrusion Materials.

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(Copies of these documents are available from https://www.astm.org.)

INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

J-STD-006 - Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications, Requirements for.

(Copies of these documents are available from https://www.ipc.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE-AS8660 - Silicone Compound NATO Code Number S-736.
SAE-AMS-H-7199 - Heat Treatment of Wrought Copper-Beryllium Alloys, Process for (Copper Alloys: Numbers C17000, C17200, C17300, C17500 and C17510.

(Copies of these documents are available from https://www.sae.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet the latter shall govern.

3.2 Material. Unless otherwise specified (see 3.1), the material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the connectors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.2.1 Brass. Brass shall be in accordance with ASTM B16/B16M, ASTM B36/B36M, ASTM B121/B121M or ASTM B124/B124M.

3.2.2 Copper-beryllium. Copper beryllium shall be in accordance with ASTM B194, ASTM B196/B196M or ASTM B197/B197M. Parts fabricated of copper beryllium shall be heat treated to condition HT after machining and forming. Heat treatment shall be in accordance with SAE-AMS-H-7199.

3.2.3 Insulating and sealing compound. Insulating and sealing compound (see 3.3.3) shall in accordance with SAE-AS8660.

3.2.4 Phosphor bronze. Phosphor bronze shall be in accordance with composition A of ASTM B139/B139M.

3.2.5 Dielectric materials. Dielectric materials shall be in accordance with the following:

3.2.5.1 Polyethylene. Polyethylene shall be in accordance with ASTM D4976.

3.2.5.2 Polystyrene. Polystyrene shall be in accordance with type E-2 of L-P-516.

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3.2.6 Silicone rubber. Silicone rubber shall be in accordance with class 2A, grade 50 or 60, of A-A-59588, except that the oil immersion test is not applicable.

3.2.7 Synthetic rubber. Synthetic rubber shall be in accordance with grade M4AA510B13F19 of ASTM D2000.

3.2.8 Silver solder. Silver solder shall be in accordance with class 1 of QQ-B-654.

3.2.8.1 Flux. Flux used while silver soldering shall be in accordance with O-F-499.

3.2.9 Soft solder. Soft solder shall be in accordance with composition Sn60 of J-STD-006.

3.3 Design and construction. Connectors shall be of the design, construction, and physical dimensions specified (see 3.1). Wherever feasible, parts having similar electrical characteristics may be combined (fabricated as a single piece) to simplify construction. Parts of unlike materials may be combined such as brass and copper beryllium, provided copper beryllium is used in the fabrication of the single-piece construction.

3.3.1 Metal parts. Unless otherwise specified (see 3.1), all metal parts shall have a silver plating of not less than .0002 inch (0.005 mm) thick, of sufficient smoothness and density to withstand the salt-spray (corrosion) test specified in 4.4.4. Silver plating shall be in accordance with ASTM B700. Dimensions of metal parts shall include the plating (see 3.1).

3.3.2 Screw threads. Screw threads shall be in accordance with FED-STD H28, and shall have the specified fit after plating (see 3.1).

3.3.3 Insulators. Insulators shall have no visible flaws or cracks. A coating of insulating and sealing compound (see 3.2.3) shall be applied to the insulators prior to assembly.

3.3.4 Gage tests for contacts (see 4.4.1.1).

3.3.4.1 Center contacts (female). Center contacts shall meet the gage tests specified in 4.4.1.1.1.

3.3.4.2 Outer contacts. Outer contacts shall meet the gage tests specified in 4.4.1.1.2.

3.3.5 Coupling nuts. After assembly, the coupling nut shall be free-finger-turning and shall not become detached from the connector body, when tested in accordance with 4.4.2.

3.4 Dielectric withstanding voltage (when applicable). When connectors are tested as specified in 4.4.3, there shall be no evidence of breakdown.

3.5 Salt spray (corrosion). When connectors are tested as specified in 4.4.4, there shall be no evidence of destructive corrosion or pitting. Destructive corrosion shall be construed as any type of corrosion, which in any way interferes with mechanical or electrical performance.

3.6 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.7 Marking. Connectors shall be marked in accordance with MIL-STD-1285, with the type designation and the manufacturer's code symbol. Marking shall be in depressed characters approximately .093 inch (2.38 mm) high, in the place specified (see 3.1).

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3.8 Workmanship. Connectors shall be processed in such a manner as to be uniform in quality and shall be free from sharp edges, burrs, except where sharp edges are required for mechanical or electrical reasons. All solder joints shall be thoroughly cleaned.

TABLE I. Component-materials inspection.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Component-materials inspection (see 4.2).

Component material	Requirement paragraph	Applicable specification
Brass	3.2.1	ASTM B16/B16M, ASTM B36/B36M, ASTM B121/B121M or ASTM B124/B124M
Copper beryllium	3.2.2	ASTM B194, ASTM B196/B196M or ASTM B197/B197M
Insulating and sealing compound	3.2.3	SAE-AS8660
Phosphor bronze	3.2.4	ASTM B139
Dielectric materials:		
Polyethylene	3.2.5.1	ASTM D4976
Polystyrene	3.2.5.2	L-P-516
Silicone rubber	3.2.6	A-A-59588
Synthetic rubber	3.2.7	ASTM D2000
Silver solder	3.2.8	QQ-B-654
Flux	3.2.8.1	O-F-499
Soft solder	3.2.9	J-STD-006

b. First article inspection (see 4.3).

4.2 Component-materials inspection. Component-materials inspection shall consist of verification that the component materials listed in table I, used in fabricating the connectors, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

4.3 First article inspection

4.3.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be made at room ambient temperature, relative humidity, and pressure.

4.3.1.1 Inspection lot. An inspection lot, shall consist of all the connectors of the same type designation, produced under essentially the same conditions, and offered for inspection at one time. However, at the discretion of the Government, an inspection lot may be composed of connectors representing a number of contracts or orders and may consist of different types of connectors in any of the following combinations:

a. Types UG-155/U and UG-287/U, latest versions.
b. Types UG-208/Uand UG-216/U, latest versions.

In combination lots, the number of samples of each different type of connector taken from the lot shall be in approximately the same proportion as the number of each type making up the lot.

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4.3.1.2 Resubmitted lots. If an inspection lot is rejected, the supplier may replace it with a new lot, rework it to correct the defects, or screen out the defective units, and submit it again for acceptance inspection. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots.

4.3.1.3 First article inspection. First article inspection shall consist of the examinations and test specified in table II, and shall be made on the same set of sample units, in the order shown.

TABLE II. First Article inspection.

Examination or test	Requirement paragraph	Method paragraph
Visual and mechanical examination:		
Design and construction:		4.4.1
Physical dimensions 1/	3.1 and 3.3	Table I and 4.4.1
Metal parts	3.3.1	4.4.1
Screw threads	3.3.2	4.4.1
Insulators	3.3.3	4.4.1
Gage tests for contacts	3.3.4	4.4.1.1
Coupling nuts	3.3.5	4.4.2
Marking	3.7	4.4.1
Workmanship	3.8	4.4.1
Dielectric withstanding voltage (when applicable)	3.4	4.4.3
Salt spray (corrosion)	3.5	4.4.4

1/ Only those dimensions related to piece parts (other than gage tests for contacts) shall be checked (see 4.3.1.3.3).

4.3.1.3.1 Sampling plan. Table II tests shall be performed on a production lot basis. Samples shall be selected in accordance with table III. If one or more defects are found, the lot shall be screened for that particular defect and defects removed. A new sample of parts shall be selected in accordance with table III and all group A tests again performed. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

4.3.1.3.2 Visual inspection (first article inspection). Each connector shall be visually examined for completeness, workmanship, and identification requirements. Attention shall be given to those assemblies that require a gasket to determine the condition of the gasket. Gaskets missing, twisted, buckled, kinked, or damaged in any way shall be cause for rejection. First article inspection shall be performed in any order acceptable to the Government.

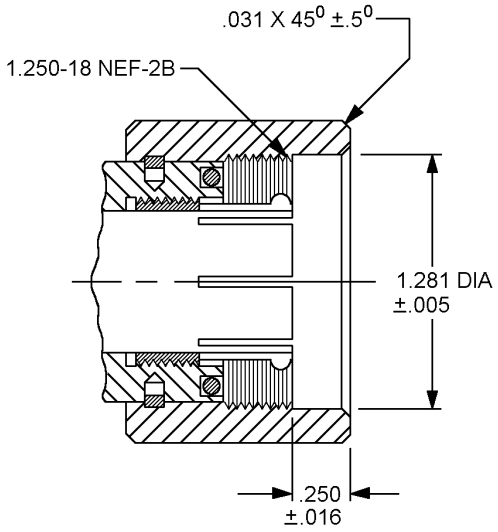
4.3.1.3.3 Physical dimensions. To facilitate inspection of the physical dimensions, the sample units shall be divided into groups of identical piece parts. Inspection of the physical dimensions shall then be performed on a group-by-group basis.

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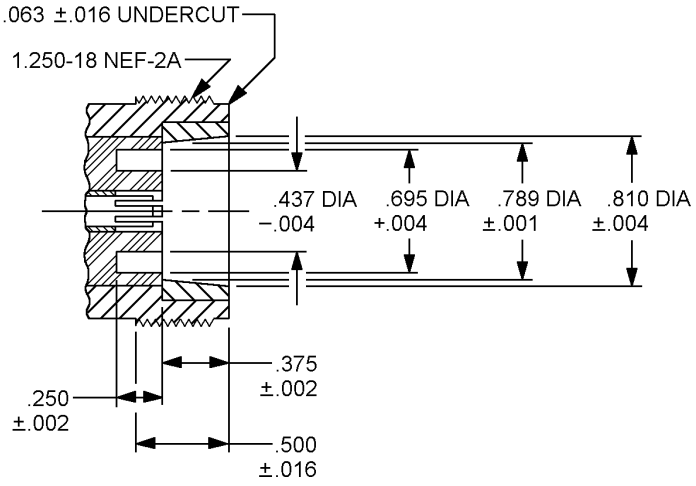
TABLE III. Inspection level.

Lot size	Visual and mechanical inspection
1 to 15	All
16 to 280	20
281 to 1,200	47
1,201 to 3,200	53
3,201 to 10,000	68
10,001 to 35,000	77
35,001 to 150,000	96
150,001 to 500,000	119
500,001 to over	143

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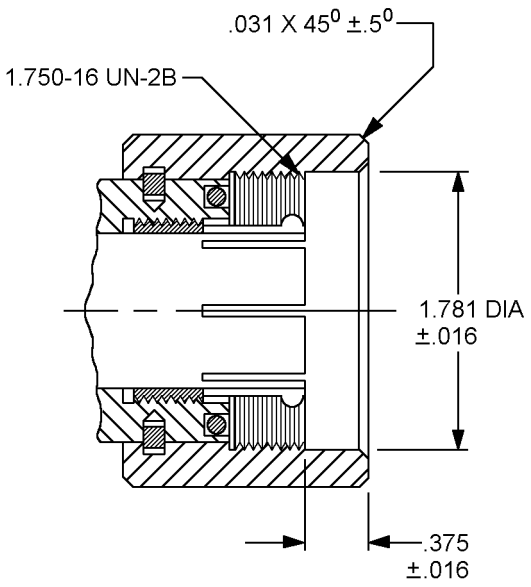
1A



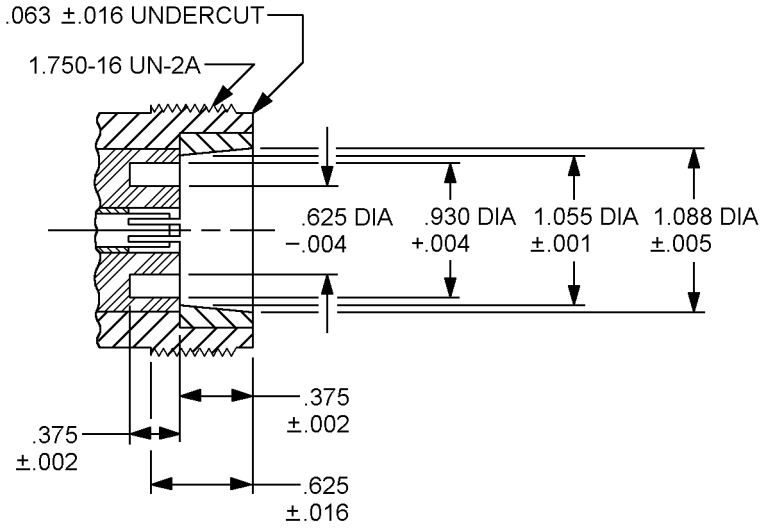
1B

FIGURE 1. Dimensional requirements for first article inspection.

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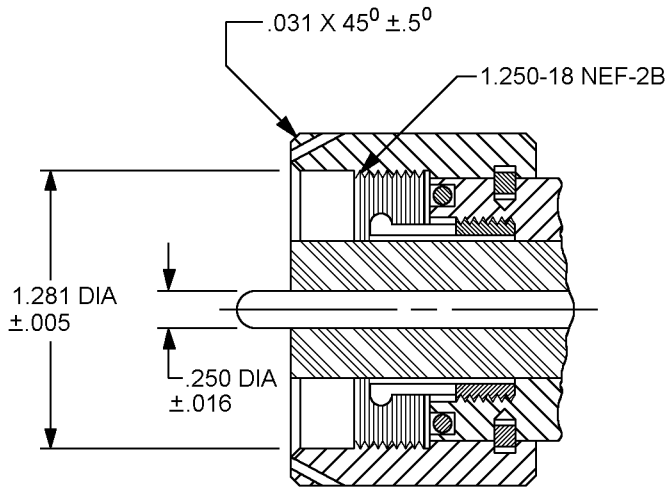
1C



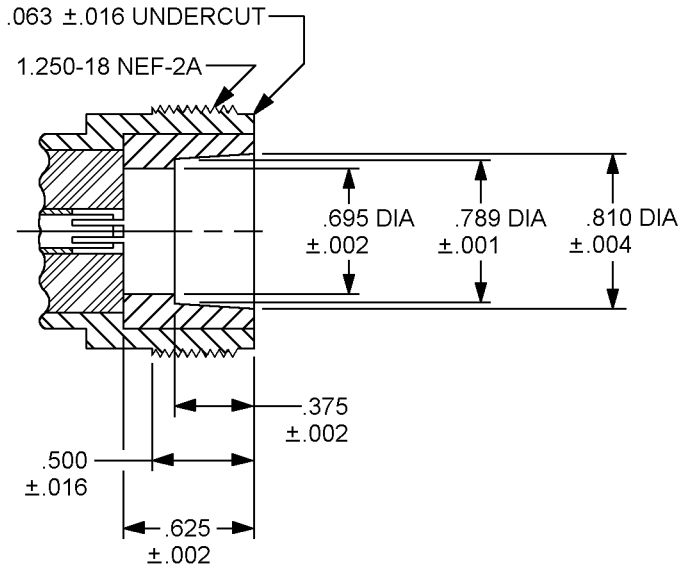
1D

FIGURE 1. Dimensional requirements for first article inspection - Continued.

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1E



1F

FIGURE 1. Dimensional requirements for first article inspection - Continued.

4.3.1.3.4 Disposition of first article sample units. Sample units which have been subjected to first article testing shall not be delivered on any contract or purchase order. The Government reserves the right to retain the sample units or to require the supplier to furnish the sample units with the first article test report

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4.3.1.3.5 Failures. One or more failures shall be cause for refusal to grant first article approval

4.3.2 Test equipment and inspection facilities. Test equipment and inspections facilities shall be of sufficient accuracy, reliability, and quantity to permit performance of the required inspection to the satisfaction of the Government.

4.4 Methods of examination and test.

4.4.1 Visual and mechanical examination. Connectors shall be examined to verify that the materials, design, construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements (see 3.1, 3.3.3, 3.6, and 3.7).

4.4.1.1 Gage tests for contacts (see 3.3.4).

4.4.1.1.1 Center contacts (female) (see 3.3.4.1).

a. Contacts mating with a .188 inch (4.77 mm) -diameter pin shall be formed in at their longitudinal centers and all contact members shall make contact with a rod .183 inch (4.65 mm) in diameter, but shall not make contact with a rod .178 inch (4.52 mm) in diameter (see 3.3.4.1).

b. Contacts mating with a .250 inch (6.35 mm) diameter pin shall be formed in at their longitudinal centers and all contact members shall make contact with a rod .254 inch (5.33 mm) in diameter, but shall not make contact with a rod .240 inch (6.10 mm) in diameter (see 3.3.4.1).

4.4.1.1.2 Outer contacts (see 3.3.4.2).

a. All fingers of the outer contacts whose nominal outside diameter is .789 inch (20.04 mm) shall make contact within .031 inch (7.87 mm) of their tip ends with a gage ring having an inside diameter of .793 inch (20.14 mm), but shall not make contact with a ring having an inside diameter of .803 inch (20.40 mm) (see 3.3.4.2).

b. All fingers of the outer contacts whose nominal outside diameter is 1.055 inches (26.80 mm) shall make contact within .031 inch (7.87 mm) of their tip ends with a gage ring having an inside diameter of 1.063 inches (27.00 mm), but shall not make contact with a ring having an inside diameter of 1.071 inches (27.20 mm) (see 3.3.4.2).

4.4.2 Coupling nuts. A pull of 100 pounds shall be applied to the coupling nut. It shall be applied in the direction away from the connector body and along the longitudinal axis (see 3.3.5).

4.4.3 Dielectric withstanding voltage (when applicable) (see 3.4). Connectors shall be tested in accordance with of MIL-STD-202-301. The following details shall apply:

a. Special preparations or conditions:
   (1) The maximum relative humidity shall be 50 percent. When facilities are not available at this test condition, connectors shall be tested at room ambient relative humidity. In case of dispute, if the test has been made at room ambient relative humidity, retest shall be made at 50 percent maximum relative humidity.
   (2) Precautions shall be taken to prevent air-gap voltage breakdowns.
   (3) The voltage shall be metered on the high side of the transformer.
b. Magnitude of test voltage - Unless otherwise specified, (see 3.1.), 10,000 volts root mean square.
c. Nature of potential - AC voltage.
d. Points of application of test voltage - Between the center contact and the body with the connector properly mated with its associated connector or a suitable mating jig.
e. Examination - After the test, connectors shall be examined for evidence of breakdown.

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4.4.4 Salt spray (corrosion) (see 3.5). Connectors shall be tested in accordance with test condition B of MIL-STD-202-101. At the conclusion of this test, the connectors shall be washed, shaken, air blasted, and then permitted to dry for 24 hours at 40°C. The connectors shall then be examined for evidence of destructive corrosion and pitting. (See 3.5.)

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Connectors covered by this specification are intended for use in radio frequency applications up to 1,000 megahertz. They are designed for use with large-size, radio frequency, coaxial cables. Their use is governed by temperature limitation of materials, and they are not recommended for use in applications where temperatures exceed 125°C.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).
c. Packaging requirements (see 5.1).
d. The supplier will not substitute for a specified material or combination of fabricated parts (see 3.3) unless he obtains approval from the Government. Evidence to substantiate his claim that such a substitution is suitable and is to be submitted with his request. Similar notification and substantiating evidence will be submitted at any later time if substitution becomes necessary or desirable. At the discretion of the Government, test sample units may be required to prove the suitability of the proposed substitute.

6.3 Cross-index. This specification covers the design of series LC connectors that were previously covered by single Service drawings. Table VII is a cross-index of the connectors and single Service drawings.

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TABLE VII. Cross-index of connectors and single Service drawings.

Type No.	MS Sheet or Bureau of Ships drawing No.
UG-154A/U	MS91604
UG-155B/U	REB49153
UG-156A/U	REB49154
UG-157B/U	REB49155
UG-208B/U	REB49156
UG-216B/U	REB49158
UG-219B/U	REB49159
UG-287B/U	MS91618
UG-352B/U	MS91610

6.4 Subject term (key word) listing.

Beryllium copper

6.5 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2022-216)

Review activities:
Army - AR,
Navy - AS, MC

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